

Questions

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Q1 - 2024 (04 Apr Shift 2)

The translational degrees of freedom (f_t) and rotational degrees of freedom (f_r) of CH_4 molecule are:

- (1) $f_t = 2$ and $f_r = 2$
- (2) $f_t = 3$ and $f_r = 3$
- (3) $f_t = 3$ and $f_r = 2$
- (4) $f_t = 2$ and $f_r = 3$

Q2 - 2024 (05 Apr Shift 1)

If the collision frequency of hydrogen molecules in a closed chamber at 27°C is Z , then the collision frequency of the same system at 127°C is :

- (1) $\frac{\sqrt{3}}{2}Z$
- (2) $\frac{2}{\sqrt{3}}Z$
- (3) $\frac{3}{4}Z$
- (4) $\frac{4}{3}Z$

Q3 - 2024 (05 Apr Shift 2)

During an adiabatic process, if the pressure of a gas is found to be proportional to the cube of its absolute temperature, then the ratio of $\frac{C_p}{C_v}$ for the gas is :

- (1) $\frac{5}{3}$
- (2) $\frac{9}{7}$
- (3) $\frac{3}{2}$
- (4) $\frac{7}{5}$

Q4 - 2024 (05 Apr Shift 2)

If n is the number density and d is the diameter of the molecule, then the average distance covered by a molecule between two successive collisions (i.e. mean free path) is represented by :

- (1) $\sqrt{2}n\pi d^2$

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(2) $\frac{1}{\sqrt{2n\pi d^2}}$

(3) $\frac{1}{\sqrt{2n\pi d^2}}$

(4) $\frac{1}{\sqrt{2n^2\pi^2 d^2}}$

Q5 - 2024 (06 Apr Shift 1)

A sample contains mixture of helium and oxygen gas. The ratio of root mean square speed of helium and oxygen in the sample, is:

(1) $\frac{1}{32}$

(2) $\frac{2\sqrt{2}}{1}$

(3) $\frac{1}{2\sqrt{2}}$

(4) $\frac{1}{4}$

Q6 - 2024 (06 Apr Shift 1)

The specific heat at constant pressure of a real gas obeying $PV^2 = RT$ equation is:

(1) $\frac{R}{3} + C_V$

(2) $C_V + R$

(3) $C_V + \frac{R}{2V}$

(4) R

Q7 - 2024 (06 Apr Shift 2)

Energy of 10 non rigid diatomic molecules at temperature T is :

(1) $70 K_B T$

(2) $35 K_B T$

(3) $\frac{7}{2} RT$

(4) $35 RT$

Q8 - 2024 (08 Apr Shift 1)

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A mixture of one mole of monoatomic gas and one mole of a diatomic gas (rigid) are kept at room temperature (27°C). The ratio of specific heat of gases at constant volume respectively is:

(1) $\frac{7}{5}$

(2) $\frac{3}{5}$

(3) $\frac{5}{3}$

(4) $\frac{3}{2}$

Q9 - 2024 (08 Apr Shift 2)

Given below are two statements :

Statement (I) : The mean free path of gas molecules is inversely proportional to square of molecular diameter.

Statement (II) : Average kinetic energy of gas molecules is directly proportional to absolute temperature of gas.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is true but Statement II is false

(2) Both Statement I and Statement II are false

(3) Both Statement I and Statement II are true

(4) Statement I is false but Statement II is true

Q10 - 2024 (09 Apr Shift 2)

The temperature of a gas is -78°C and the average translational kinetic energy of its molecules is K . The temperature at which the average translational kinetic energy of the molecules of the same gas becomes $2K$ is

(1) 127°C

(2) 117°C

(3) -39°C

(4) -78°C

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Answer Key

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Solutions

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Q1

Since CH_4 is polyatomic Non-LinearD.O.F of CH_4

T. DOF = 3

RDOF = 3

Q2

Assuming mean free path constant.

$$f \propto V \propto \sqrt{T}$$

$$\frac{f_1}{f_2} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{300}{400}}$$

$$f_2 = \sqrt{\frac{4}{3}} = f_1 = \frac{2}{\sqrt{3}} Z$$

Q3

$$P \propto T^3$$

$$PT^{-3} = \text{constant}$$

$$\therefore \frac{PV}{T} = nR = \text{constant from ideal gas equation}$$

$$(P)(PV)^{-3} = \text{constant}$$

$$P^{-2} V^{-3} = \text{constant} \dots (i)$$

 $\therefore$ Process equation for adiabatic process is

$$PV^y = \text{constant} \dots (ii)$$

Comparing equation (1) and (2)

$$\frac{C_P}{C_V} = \gamma = \frac{3}{2}$$

Q4

n = number of molecule per unit volume

d = diameter of the molecule

$$\lambda = \frac{1}{\sqrt{2} \pi d^2 n}$$

(By Theory)

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Solutions

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Q5

$$V_{ms} = \sqrt{\frac{3RT}{M_w}}$$

$$\Rightarrow \frac{V_{O_2}}{V_{He}} = \sqrt{\frac{M_{w,He}}{M_{w,O_2}}}$$

$$= \sqrt{\frac{4}{32}} = \frac{1}{2\sqrt{2}}$$

$$\frac{V_{He}}{V_{O_2}} = \frac{2\sqrt{2}}{1}$$

Q6

$$dQ = du + dW$$

$$CdT = C_vdT + PdV$$

$$\therefore PV^2 = RT$$

$$P = \text{constant}$$

$$P(2VdV) = RdT$$

$$PdV = \frac{RdT}{2V}$$

Put in equation (1)

$$C = C_v + \frac{R}{2V}$$

Q7

$$\text{Degree of freedom } (f) = 5 + 2(3N - 5)$$

$$f = 5 + 2(3 \times 2 - 1) = 7$$

$$\text{energy of one molecule} = \frac{f}{2} K_B T$$

energy of 10 molecules

$$= 10 \left(\frac{f}{2} K_B T \right) = 10 \left(\frac{7}{2} K_B T \right) = 35 K_B T$$

Q8

$$\frac{(C_v)_{\text{mono}}}{(C_v)_{\text{dia}}} = \frac{\frac{3}{2}R}{\frac{5}{2}R} = \frac{3}{5}$$

Q9

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Solutions

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$$\lambda = \frac{RT}{\sqrt{2}\pi d^2 N_A P}$$

$$KE = \frac{f}{2} nRT$$

Q10

$$K.E = \frac{nf_1 RT}{2}$$

$$T_i = -78^\circ\text{C} \rightarrow 273 + [-78^\circ\text{C}] = 195\text{ K}$$

$$K.E \propto T$$

To double the K.E energy temp also become double

$$T_f = 390\text{ K}$$

$$T_f = 117^\circ\text{C}$$

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